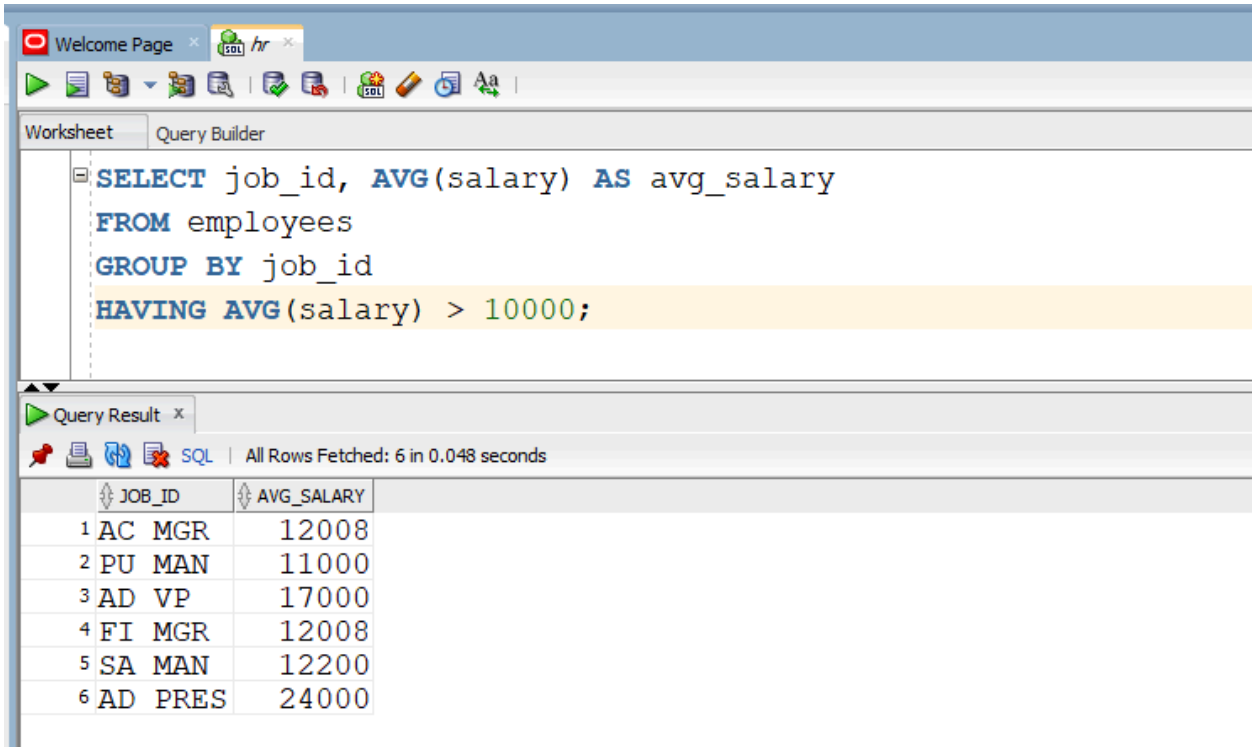
DBS LAB 6

IN LAB TASK

TASK 1:



TASK 2:

Welcome Page | hr

Worksheet | Query Builder

```sql
SELECT department_id, COUNT(*) AS num_employees
FROM employees
WHERE salary > 2000
GROUP BY department_id
HAVING COUNT(*) > 2;
```

Query Result

SQL | All Rows Fetched: 6 in 0.09 seconds

	DEPARTMENT_ID	NUM_EMPLOYEES
1	100	6
2	30	6
3	90	3
4	50	45
5	80	34
6	60	5

TASK 3:

Welcome Page | hr

Worksheet | Query Builder

```sql
SELECT department_name
FROM (
    SELECT d.department_name
    FROM departments d
    JOIN employees e
    ON d.department_id = e.department_id
    ORDER BY e.hire_date DESC
)
WHERE ROWNUM = 1;
```

Query Result

SQL | All Rows Fetched: 1 in 0.048 seconds

	DEPARTMENT_NAME
1	Sales

TASK 4:

Welcome Page | hr

Worksheet | Query Builder

```
CREATE INDEX emp_index
ON employees(employee_id);

CREATE INDEX job_hist_index
ON job_history(employee_id);

SELECT *
FROM employees
WHERE employee_id NOT IN (
    SELECT employee_id FROM job_history
);
```

Query Result

SQL | Fetched 50 rows in 0.089 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
3	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
4	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
5	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
8	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
9	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
10	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
11	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
12	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
13	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100	(null)	114	30
14	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900	(null)	114	30
15	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
16	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(null)	114	30
17	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500	(null)	114	30
18	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
19	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50

Activate Windows
Go to Settings to activate

TASK 5:

Welcome Page | hr

Worksheet | Query Builder

```
SELECT department_id, MAX(salary) AS highest_salary
FROM employees
GROUP BY department_id
HAVING MAX(salary) >= 2000
ORDER BY highest_salary ASC;
```

Query Result

SQL | All Rows Fetched: 11 in 0.024 seconds

	DEPARTMENT_ID	HIGHEST_SALARY
1	10	4400
2	40	6500
3	(null)	7000
4	50	8200
5	60	9000
6	70	10000
7	30	11000
8	100	12008
9	110	12008
10	80	14000
11	90	24000

TASK 6:

Welcome Page | hr

Worksheet | Query Builder

```
SELECT *
FROM (
    SELECT department_id, AVG(salary) AS avg_salary
    FROM employees
    GROUP BY department_id
    HAVING AVG(salary) > 6000
)
WHERE ROWNUM <= 3;
```

Query Result

SQL | All Rows Fetched: 3 in 0.005 seconds

	DEPARTMENT_ID	AVG_SALARY
1	100	8601.33333333333333333333333333333333333
2	(null)	7000
3	90	19333.33333333333333333333333333333333333

TASK 7:

Welcome Page | hr

Worksheet | Query Builder

```
SELECT last_name, job_id
FROM employees
WHERE job_id = (
    SELECT job_id
    FROM employees
    WHERE employee_id = 150
);
```

Query Result

SQL | All Rows Fetched: 30 in 0.013 seconds

	LAST_NAME	JOB_ID
1	Tucker	SA_REP
2	Bernstein	SA_REP
3	Hall	SA_REP
4	Olsen	SA_REP
5	Cambrault	SA_REP
6	Tuvault	SA_REP
7	King	SA_REP
8	Sully	SA_REP
9	McEwen	SA_REP
10	Smith	SA_REP
11	Doran	SA_REP
12	Sewall	SA_REP
13	Vishney	SA_REP
14	Greene	SA_REP
15	Marvins	SA_REP
16	Lee	SA_REP
17	Ande	SA_REP
18	Banda	SA_REP
19	Ozer	SA_REP
20	Bloom	SA_REP
21	Fox	SA_REP
22	Smith	SA_REP
23	Bates	SA_REP
24	Kumar	SA_REP

Activate
Go to Sett

TASK 8:

```
CREATE TABLE Job_History1
AS SELECT * FROM job_history WHERE 1=2;
```

```
Error at Command Line : 4 Column : 13
Error report -
SQL Error: ORA-00942: table or view does not exist
00942. 00000 -  "table or view does not exist"
*Cause:
*Action:

Error starting at line : 4 in command -
INSERT INTO Job_History1
SELECT *
FROM hr.job_history
WHERE end_date = '19-DEC-07'
Error at Command Line : 4 Column : 13
Error report -
SQL Error: ORA-00942: table or view does not exist
00942. 00000 -  "table or view does not exist"
*Cause:
*Action:

Table JOB_HISTORY1 created.
```

Welcome Page | hr

Worksheet | Query Builder

```
INSERT INTO Job_History1
SELECT *
FROM hr.job_history
WHERE end_date = '19-DEC-07';
```

Query Result | Script Output

Task completed in 0.057 seconds

```
00942. 00000 -  "table or view does not exist"
*Cause:
*Action:

Error starting at line : 4 in command -
INSERT INTO Job_History1
SELECT *
FROM hr.job_history
WHERE end_date = '19-DEC-07'
Error at Command Line : 4 Column : 13
Error report -
SQL Error: ORA-00942: table or view does not exist
00942. 00000 -  "table or view does not exist"
*Cause:
*Action:

Table JOB_HISTORY1 created.


0 rows inserted.
```

TASK 9:

Welcome Page | hr

Worksheet | Query Builder

```
INSERT INTO Job_History1 VALUES (101, '01-JAN-05','01-JAN-06','IT_PROG',90);
INSERT INTO Job_History1 VALUES (102, '01-FEB-05','01-FEB-06','AC_ACCOUNT',100);
INSERT INTO Job_History1 VALUES (103, '01-MAR-05','01-MAR-06','SA_REP',80);
INSERT INTO Job_History1 VALUES (104, '01-APR-05','01-APR-06','AC_ACCOUNT',60);
INSERT INTO Job_History1 VALUES (105, '01-MAY-05','01-MAY-06','IT_PROG',50);

DELETE FROM Job_History1
WHERE job_id = 'AC_ACCOUNT';
```

Query Result | Script Output

Task completed in 0.057 seconds

```
FROM hr.job_history
WHERE end_date = '19-DEC-07'
Error at Command Line : 4 Column : 13
Error report -
SQL Error: ORA-00942: table or view does not exist
00942. 00000 -  "table or view does not exist"
*Cause:
*Action:

Table JOB_HISTORY1 created.


0 rows inserted.


0 rows deleted.
```

TASK 10:

```
DELETE FROM job_history
WHERE department_id = (
    SELECT department_id
    FROM departments
    WHERE department_name = 'IT'
);
```

Task completed in 0.122 seconds

```
1 row inserted.


Error starting at line : 1 in command -
INSERT INTO Job_History1 VALUES (101, '01-JAN-05','01-JAN-06','IT_PROG',90)
INSERT INTO Job_History1 VALUES (102, '01-FEB-05','01-FEB-06','AC_ACCOUNT',100)
INSERT INTO Job_History1 VALUES (103, '01-MAR-05','01-MAR-06','SA_REP',80)
INSERT INTO Job_History1 VALUES (104, '01-APR-05','01-APR-06','AC_ACCOUNT',60)
INSERT INTO Job_History1 VALUES (105, '01-MAY-05','01-MAY-06','IT_PROG',50)
Error at Command Line : 2 Column : 8
Error report -
SQL Error: ORA-00933: SQL command not properly ended
00933. 00000 -  "SQL command not properly ended"
*Cause:
*Action:

1 row deleted.
```

POST LAB TASK

TASK 1:

Welcome Page | hr

Worksheet | Query Builder

```
SELECT first_name, salary
FROM employees e
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
);
```

Script Output | Query Result

All Rows Fetched: 37 in 0.035 seconds

	FIRST_NAME	SALARY
1	Steven	24000
2	Alexander	9000
3	Bruce	6000
4	Nancy	12008
5	Daniel	9000
6	Den	11000
7	Matthew	8000
8	Adam	8200
9	Payam	7900
10	Shanta	6500
11	Kevin	5800
12	Renske	3600
13	Trenna	3500
14	John	14000
15	Karen	13500
16	Alberto	12000
17	Gerald	11000
18	Eleni	10500
19	Peter	10000
20	David	9500
21	Peter	9000
22	Janette	10000
23	Patrick	9500

Activ
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TASK 2:

Welcome Page | hr

Worksheet | Query Builder

```sql
SELECT job_id, MIN(salary) AS min_salary
FROM employees
GROUP BY job_id
HAVING MIN(salary) >= 1000
ORDER BY min_salary ASC;
```

Script Output | Query Result

All Rows Fetched: 17 in 0.071 seconds

	JOB_ID	MIN_SALARY
1	ST_CLERK	2100
2	PU_CLERK	2500
3	SH_CLERK	2500
4	IT_PROG	4200
5	AD_ASST	4400
6	ST_MAN	5800
7	SA_REP	6100
8	HR_REP	6500
9	FI_ACCOUNT	6900
10	AC_ACCOUNT	8300
11	PR_REP	10000
12	SA_MAN	10500
13	PU_MAN	11000
14	FI_MGR	12008
15	AC_MGR	12008
16	AD_VP	17000
17	AD_PRES	24000

TASK 3:

Welcome Page | hr

Worksheet | Query Builder

```
SELECT first_name, department_id
FROM employees
WHERE department_id = (
    SELECT department_id
    FROM employees
    WHERE employee_id = 140
);
```

Script Output | Query Result

SQL | All Rows Fetched: 45 in 0.066 seconds

	FIRST_NAME	DEPARTMENT_ID
1	Matthew	50
2	Adam	50
3	Payam	50
4	Shanta	50
5	Kevin	50
6	Julia	50
7	Irene	50
8	James	50
9	Steven	50
10	Laura	50
11	Mozhe	50
12	James	50
13	TJ	50
14	Jason	50
15	Michael	50
16	Ki	50
17	Hazel	50
18	Renske	50
19	Stephen	50
20	John	50
21	Joshua	50
22	Trenna	50
23	Curtis	50

TASK 4:

Welcome Page | hr

Worksheet | Query Builder

```
CREATE TABLE employees_copy AS
SELECT * FROM employees;
```

Script Output | Query Result

Task completed in 0.078 seconds

```
Error starting at line : 1 in command -
INSERT INTO Job_History1 VALUES (101, '01-JAN-05','01-JAN-06','IT_PROG',90)
INSERT INTO Job_History1 VALUES (102, '01-FEB-05','01-FEB-06','AC_ACCOUNT',100)
INSERT INTO Job_History1 VALUES (103, '01-MAR-05','01-MAR-06','SA_REP',80)
INSERT INTO Job_History1 VALUES (104, '01-APR-05','01-APR-06','AC_ACCOUNT',60)
INSERT INTO Job_History1 VALUES (105, '01-MAY-05','01-MAY-06','IT_PROG',50)
Error at Command Line : 2 Column : 8
Error report -
SQL Error: ORA-00933: SQL command not properly ended
00933. 00000 -  "SQL command not properly ended"
*Cause:
*Action:

1 row deleted.


Table EMPLOYEES_COPY created.
```

Welcome Page | hr

Worksheet | Query Builder

```
UPDATE employees_copy
SET salary = salary * 1.12
WHERE salary BETWEEN 5000 AND 10000;
```

Script Output | Query Result

Task completed in 0.123 seconds

```
Error starting at line : 1 in command -
INSERT INTO Job_History1 VALUES (101, '01-JAN-05','01-JAN-06','IT_PROG',90)
INSERT INTO Job_History1 VALUES (102, '01-FEB-05','01-FEB-06','AC_ACCOUNT',100)
INSERT INTO Job_History1 VALUES (103, '01-MAR-05','01-MAR-06','SA_REP',80)
INSERT INTO Job_History1 VALUES (104, '01-APR-05','01-APR-06','AC_ACCOUNT',60)
INSERT INTO Job_History1 VALUES (105, '01-MAY-05','01-MAY-06','IT_PROG',50)
Error at Command Line : 2 Column : 8
Error report -
SQL Error: ORA-00933: SQL command not properly ended
00933. 00000 -  "SQL command not properly ended"
*Cause:
*Action:

1 row deleted.


Table EMPLOYEES_COPY created.


42 rows updated.
```

TASK 5:

```
SELECT *
FROM (
    SELECT first_name, salary
    FROM employees
    ORDER BY salary DESC
)
WHERE ROWNUM <= (
    SELECT COUNT(*)*0.1 FROM employees
);
```

All Rows Fetched: 10 in 0.006 seconds

	FIRST_NAME	SALARY
1	Steven	24000
2	Neena	17000
3	Lex	17000
4	John	14000
5	Karen	13500
6	Nancy	12008
7	Shelley	12008
8	Alberto	12000
9	Lisa	11500
10	Den	11000

TASK 6:

```
SELECT location_id, COUNT(department_id)
FROM departments
GROUP BY location_id
HAVING COUNT(department_id) > 1;
```

All Rows Fetched: 1 in 0.069 seconds

	LOCATION_ID	COUNT(DEPARTMENT_ID)
1	1700	21

TASK 7:

Welcome Page | hr

Worksheet | Query Builder

```sql
SELECT job_id, MIN(salary) AS min_salary
FROM employees
WHERE job_id IS NOT NULL
GROUP BY job_id
HAVING MIN(salary) != 1500
ORDER BY min_salary DESC;
```

Script Output | Query Result

SQL | All Rows Fetched: 17 in 0.008 seconds

	JOB_ID	MIN_SALARY
1	AD PRES	24000
2	AD VP	17000
3	AC MGR	12008
4	FI MGR	12008
5	PU MAN	11000
6	SA MAN	10500
7	PR REP	10000
8	AC ACCOUNT	8300
9	FI ACCOUNT	6900
10	HR REP	6500
11	SA REP	6100
12	ST MAN	5800
13	AD ASST	4400
14	IT PROG	4200
15	SH CLERK	2500
16	PU CLERK	2500
17	ST CLERK	2100

TASK 8:

```sql
SELECT first_name, department_id
FROM employees
WHERE department_id = (
    SELECT department_id
    FROM employees
    WHERE employee_id = 130
);
```

All Rows Fetched: 45 in 0.004 seconds

	FIRST_NAME	DEPARTMENT_ID
1	Matthew	50
2	Adam	50
3	Payam	50
4	Shanta	50
5	Kevin	50
6	Julia	50
7	Irene	50
8	James	50
9	Steven	50
10	Laura	50
11	Mozhe	50
12	James	50
13	TJ	50
14	Jason	50
15	Michael	50
16	Ki	50
17	Hazel	50
18	Renske	50
19	Stephen	50
20	John	50
21	Joshua	50
22	Trenna	50
23	Curtis	50
24	Randall	50
25	Peter	50

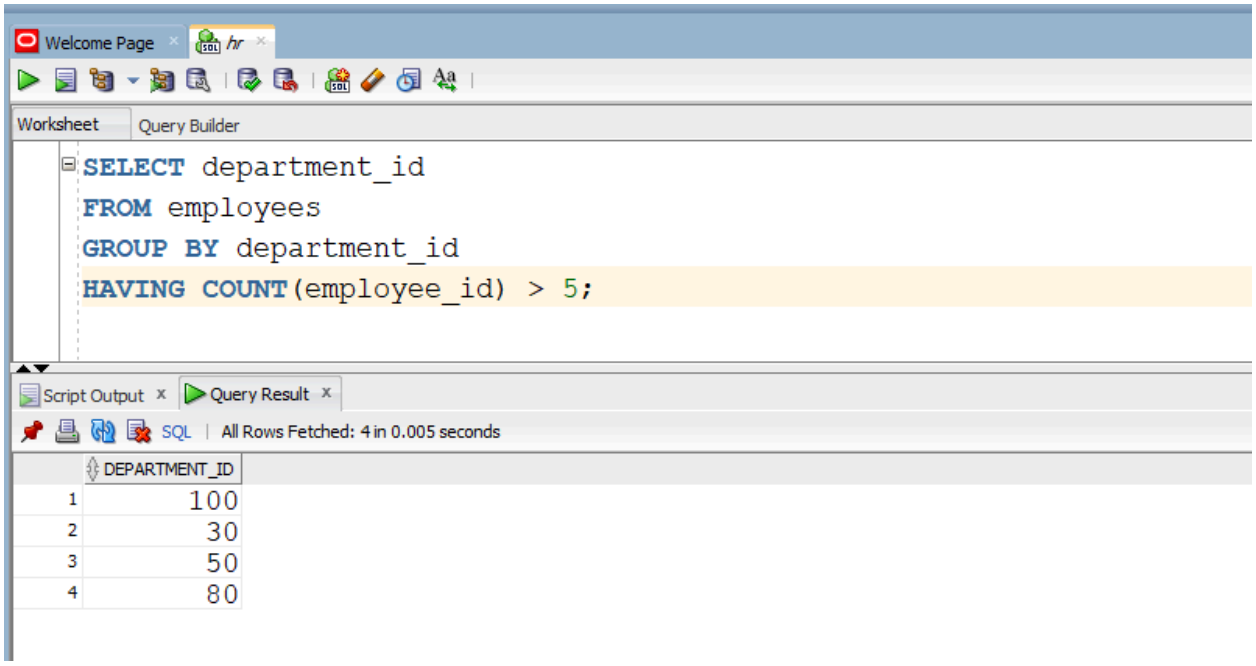
TASK 9:

```sql
SELECT *
FROM employees
WHERE department_id != 30
AND salary > ALL (
    SELECT salary
    FROM employees
    WHERE department_id = 30
);
```

All Rows Fetched: 9 in 0.074 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
2	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
3	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101	110
4	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
5	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
6	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
7	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
8	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
9	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90

TASK 10:


Welcome Page
hr
Worksheet
Query Builder
SELECT department_id
FROM employees
GROUP BY department_id
HAVING COUNT(employee_id) > 5;
Script Output
Query Result
SQL
All Rows Fetched: 4 in 0.005 seconds
DEPARTMENT_ID
1 100
2 30
3 50
4 80